Week 12: 16–20 November 2026

Subject / الموضوع: Science **Unit Title / عنوان الوحدة: Forces, Motion and Energy – القوى والحركة والطاق** **Grade / الصف: 8**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Session 1 الحصة الأولى Force Pairs	**DOK 3 – Explain with evidence:** I can use evidence from investigations to explain Newton's third law and analyze the action–reaction force pairs acting in real situations.	3.3 Guide processing with explicit steps, chunking, and interactive models. 8.3 Foster collaboration and community with structured group roles and supports. 4.1 Provide alternatives for physical action and methods of response.	Kindly check Toddle/Teams for the lesson PPT and practice sheet.
Session 2 الحصة الثانية Gravitational Force	**DOK 3 – Analyze:** I can analyze how mass and distance affect the strength of the gravitational force between objects, and use evidence to explain the difference between mass and weight.	7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Kindly check Toddle/Teams for the lesson PPT and practice sheet.
Session 3 الحصة الثالثة Kinetic and Potential Energy	**DOK 3 – Analyze data:** I can analyze data to explain how kinetic energy depends on mass and speed and how potential energy depends on height, and describe how energy changes from one form to the other.	7.1 Increase engagement by offering choices and autonomy. 5.1 Provide multiple communication media (text, speech, visuals, models). 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	Kindly check Toddle/Teams for the lesson PPT and practice sheet.
Session 4 الحصة الرابعة Conservation of Energy	**DOK 3 – Analyze:** I can analyze the energy changes in a system to explain, with evidence, that energy is conserved, and account for energy transferred as heat or sound.	3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 8.3 Foster collaboration and community with structured group roles and supports. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	Kindly check Toddle/Teams for the lesson PPT and practice sheet.

Homework **الواجبات**			
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	This week students study force pairs (Session 1), gravity (Session 2), kinetic and potential energy (Session 3) and the conservation of energy (Session 4). Students will use calculators in every session. Please encourage your child to practise the formulas at home: $F = m \times a$, $W = m \times g$, $KE = ½ \times m \times v^2$ and $GPE = m \times g \times h$.		

Week 11 الأسبوع 11

Technology Integration دمج التكنولوجيا	Videos and online simulations of motion, forces and energy, and Toddle for practice sheets and feedback.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics (rearranging formulas, graphs and units) and Physical and Health Education (forces and energy in sport).
External Assessment Integration دمج التقييمات الخارجية	Aligned to NGSS MS-PS2-1, MS-PS2-4, MS-PS3-1, MS-PS3-2 and MS-PS3-5, and MAP Growth Science (Physical Sciences).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking (analysing data and graphs), collaboration (group investigations) and communication (explaining answers with scientific vocabulary and units).
Standards المعايير	• **MS-PS2-1:** Apply Newton's third law to design a solution to a problem involving the motion of two colliding objects. • **MS-PS2-4:** Construct and present arguments using evidence to support the claim that gravitational interactions are attractive and depend on the masses of interacting objects. • **MS-PS3-1:** Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass of an object and to the speed of an object. • **MS-PS3-2:** Develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts of potential energy are stored in the system. • **MS-PS3-5:** Construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Inquiry:** How do forces and energy explain the motion of objects, and why is energy never lost? **TOK:** We cannot see energy; we only observe what it does. How do scientists know that energy is conserved?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Session 1 – Force Pairs** • **MUST (DOK 1):** I can state Newton's third law and identify the action and reaction forces in a picture. • **SHOULD (DOK 2):** I can explain, using examples such as a rocket, swimming or walking, that the two forces in a pair are equal in size, opposite in direction, and act on different objects. • **COULD (DOK 3):** I can justify why a force pair does not cancel out, and predict how two objects of different mass move after a collision using $F = m \times a$. **Session 2 – Gravitational Force** • **MUST (DOK 1):** I can define gravitational force, mass and weight, and state that mass is measured in kg and weight in N.

	• **SHOULD (DOK 2):** I can calculate weight using W = m × g and explain how mass and distance change the gravitational force. • **COULD (DOK 3):** I can compare an object's weight on Earth, the Moon and Mars, and justify the differences using data. **Session 3 – Kinetic and Potential Energy** • **MUST (DOK 1):** I can define kinetic energy and gravitational potential energy and identify examples of each. • **SHOULD (DOK 2):** I can calculate KE = ½ × m × v^2 and GPE = m × g × h, and explain the energy changes on a roller coaster. • **COULD (DOK 3):** I can use data to justify why speed has a greater effect on kinetic energy than mass, and analyze the energy changes in a real system. **Session 4 – Conservation of Energy** • **MUST (DOK 1):** I can state the law of conservation of energy and identify the energy changes in a system, such as a swinging pendulum. • **SHOULD (DOK 2):** I can explain, using energy bar charts, how kinetic and potential energy change while the total energy stays the same. • **COULD (DOK 3):** I can calculate KE and GPE at different points to show that the total energy stays the same, and explain where the "lost" energy goes (heat and sound from friction).
National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء Global Understanding التفاهم العالمي Arabic Language اللغة العربية
	Area:المجال Pride in UAE achievements in science, transport and space exploration.
	Implementation:التطبيق • Session 1: students explain the force pairs that lift a rocket, linked to the launch of the Emirates Mars Mission (Hope Probe). • Session 2: students compare weight on Earth, the Moon and Mars, linked to the Hope Probe and the Rashid lunar rover. • Sessions 3–4: students analyze the energy changes on Formula Rossa at Ferrari World Abu Dhabi and show that the total energy is conserved. • Arabic key terms: أزواج القوى (force pairs), القوة (force), الجاذبية (gravity), الطاقة الحركية (kinetic energy), طاقة الوضع (potential energy), حفظ الطاقة (conservation of energy).